

Systems of Linear Equations (AA HL)



Solving a 2x2 system by elimination

- 1 Solve the system $x + 2y = 8$, $3x - y = 3$ by elimination.
 - 2 Solve the system $3x + 2y = 16$, $x - y = 2$ by elimination.
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Solving a 2x2 system by substitution

- 3 Solve the system $x - 3y = -5$, $2x + y = 4$ by substitution.
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A system with no solution

- 4 Show that the system $x + y = 4$, $2x + 2y = 9$ has no solution.
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A system with infinitely many solutions

- 5 Show that the system $2x - 4y = 6$, $x - 2y = 3$ has infinitely many solutions, and write the general solution using a parameter t .
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Solving a 3x3 system by elimination

- 6 Solve the system $x + y + z = 6$, $x - y + 2z = 5$, $2x + y - z = 1$.
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A 3x3 system with no solution

- 7 Explain why the system $x + y + z = 4$, $2x + 2y + 2z = 10$, $x - y + z = 2$ has no solution.
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A 3x3 system with infinitely many solutions

- 8 Show that the system $x + y + z = 6$, $x - y + z = 2$, $2x + 2z = 8$ has infinitely many solutions, and write the general solution using a parameter t .
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Classifying a system without fully solving it

- 9 Without solving, determine whether $x + 2y = 5$, $2x + 4y = 9$ has a unique solution, no solution, or infinitely many solutions. Justify your answer using the coefficients.
- 10 Without solving, determine whether $x - y = 3$, $2x - 2y = 6$ has a unique solution, no solution, or infinitely many solutions. Justify your answer using the coefficients.
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Geometric interpretation: where a line of intersection meets a third plane

- 11 The planes $x + y + z = 4$ and $x - y + z = 2$ intersect in the line $(3 - t, 1, t)$. Find the point where this line meets the plane $2x + z = 9$.
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Application: fitting a quadratic through three points

- 12 Find a , b , c such that the parabola $y = ax^2 + bx + c$ passes through $(1, 6)$, $(2, 11)$, and $(3, 18)$, by setting up and solving a system of three linear equations.
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Application: a cost word problem

- 13 3 adult tickets and 2 child tickets cost 58 dollars in total. 2 adult tickets and 5 child tickets cost 68 dollars in total. Find the price of an adult ticket and the price of a child ticket by setting up and solving a system of two linear equations.
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Application: a three-numbers word problem

- 14 The sum of three numbers is 18. The second number is twice the first, and the third number is three times the first. Find the three numbers by setting up and solving a system of three linear equations.
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Solving a system with decimal coefficients

- 15 Solve the system $0.4x + 0.3y = 2.5$, $0.2x - 0.1y = 0.5$.
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Solving a 2x2 system using Cramer's Rule

- 16 Use Cramer's Rule to solve $2x + 3y = 8$, $x - y = -1$.

A harder 3x3 system

- 17 Solve $x + 2y - z = 3$, $2x - y + z = 7$, $x + y + 2z = 8$.
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Finding a parameter for infinitely many solutions

- 18 Find the value of k for which the system $3x + ky = 7$, $6x + 4y = 14$ has infinitely many solutions.
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Finding a parameter for no solution

- 19 Find the value of k for which the system $x + 3y = 4$, $2x + ky = 9$ has no solution.
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Geometric interpretation: two parallel planes

- 20 The planes $x + y + z = 1$, $x + y + z = 4$, and $2x - y + z = 3$ have no point in common. Explain why, without needing the third equation.
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Application: a traffic-flow word problem

- 21 At a road network, flow conservation at two junctions gives $2x - y = 30$ and $x + y = 90$, where x and y are traffic flows in vehicles per hour. Find x and y .
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Application: balancing a chemical equation

- 22 Balance the combustion equation $aC_3H_8 + bO_2 \rightarrow cCO_2 + dH_2O$ by setting $a = 1$ and using conservation of carbon, hydrogen, and oxygen atoms to set up and solve a system of linear equations for b , c , d .
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Application: a three-way investment word problem

- 23 12000 dollars is invested across three accounts paying 3%, 4%, and 5% annual interest. The amount in the 4% account is twice the amount in the 3% account, and the total annual interest earned is 500 dollars. Find the amount invested in each account.
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Application: recognizing an inconsistent price system

- 24** A shop sells only pens (price p dollars each) and notebooks (price n dollars each). One customer buys 3 pens and 2 notebooks for 16 dollars. Another buys 6 pens and 4 notebooks for 35 dollars. Explain why these two totals cannot both be correct.